

LITERATURE-IMPLIED ANSWER TO THE HILBERT L_2 POWER-FUNCTION PROBLEM

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STATUS

This is a lightweight literature-implied answer packet. It records that Open Problem 1 in arXiv:1011.3682 is answered affirmatively by the later Hilbert L_2 sampling theorem of Krieg–Ullrich, arXiv:1905.02516.

SOURCE QUESTION

The source paper is Erich Novak and Henryk Wozniakowski, *On the power of function values for the approximation problem in various settings*, arXiv:1011.3682.

In Section 2.1, after Theorem 1, the paper asks:

Suppose $r > \frac{1}{2}$. Is it true that $\ell^{\text{wor-H}}(r, 2) = 1$?

Equivalently, in the source notation, whenever the n -th minimal worst-case error using arbitrary linear information has polynomial rate $r > 1/2$ for a Hilbert-space approximation problem measured in L_2 , do function values have the same polynomial power?

SUPPORTING RESULT

The supporting paper is David Krieg and Mario Ullrich, *Function values are enough for L_2 -approximation*, arXiv:1905.02516, published in *Foundations of Computational Mathematics* 21 (2021), 1141–1151.

Their main theorem gives constants $C, c > 0$ and $k_n \geq cn/\log(n+1)$ such that, for every reproducing kernel Hilbert space $H \hookrightarrow L_2$,

$$e_n(H)^2 \leq \frac{C}{k_n} \sum_{j \geq k_n} a_j(H)^2,$$

where e_n are sampling numbers, i.e. errors using function values, and a_n are approximation numbers, i.e. errors using arbitrary linear information. Their Corollary 1 states that if

$$a_n(H) \lesssim n^{-s} \log^\alpha(n) \quad (s > 1/2),$$

then

$$e_n(H) \lesssim n^{-s} \log^{\alpha+s}(n),$$

and in particular $\text{ord}(e_n) = \text{ord}(a_n)$.

Date: 2026-07-18.

IMPLICATION FOR ARXIV:1011.3682

Fix $r > 1/2$. Suppose the all-linear worst-case Hilbert L_2 error has source rate r , i.e. $\text{ord}(a_n) = r$. For every $s < r$, the sequence a_n is bounded by n^{-s} up to a constant, so Krieg–Ullrich’s corollary gives $\text{ord}(e_n) \geq s$. Letting $s \uparrow r$ yields $\text{ord}(e_n) \geq r$. Since function values are a restricted class of linear information, $a_n \leq e_n$, hence $\text{ord}(e_n) \leq \text{ord}(a_n) = r$. Therefore

$$\text{ord}(e_n) = \text{ord}(a_n) = r.$$

In the power-function notation of arXiv:1011.3682, this is exactly

$$\ell^{\text{wor-H}}(r, 2) = 1 \quad \text{for every } r > \frac{1}{2}.$$

SCOPE

This packet answers only Open Problem 1 of the source paper. It does not settle the source’s open problems for $p \neq 2$, Banach worst-case power functions, randomized settings, average-case $p \neq 2$, or the fixed- n supremum ratio problem.

PROVENANCE

The supporting paper explicitly answers the equivalent open question about the equality of polynomial orders for Hilbert L_2 sampling and arbitrary linear information. In the TeX source checked for this packet, it does not appear to cite arXiv:1011.3682 directly; the connection to the source Open Problem 1 is the power-function reformulation above.

REFERENCES

- [1] E. Novak and H. Wozniakowski, *On the power of function values for the approximation problem in various settings*, arXiv:1011.3682.
- [2] D. Krieg and M. Ullrich, *Function values are enough for L_2 -approximation*, arXiv:1905.02516; Found. Comput. Math. 21 (2021), 1141–1151.